MAE 184 Flight Simulation Techniques
Homework #1 Solution

Problem 1.1

A flight test technique used to estimate the drag coefficient of early airships involved accelerating the vehicle up to its maximum forward speed, cutting the engines, and then measuring how the vehicle speed slowed over time. Use Euler integration to find the vehicle speed after it has coasted without engine power for 30 seconds. Use an integration time step of 0.1 seconds and assume an initial airspeed of 70 ft/s.

$$m\frac{dV}{dt}=-\frac{1}{2}\rho V^2 \nabla^{2/3} C_D$$

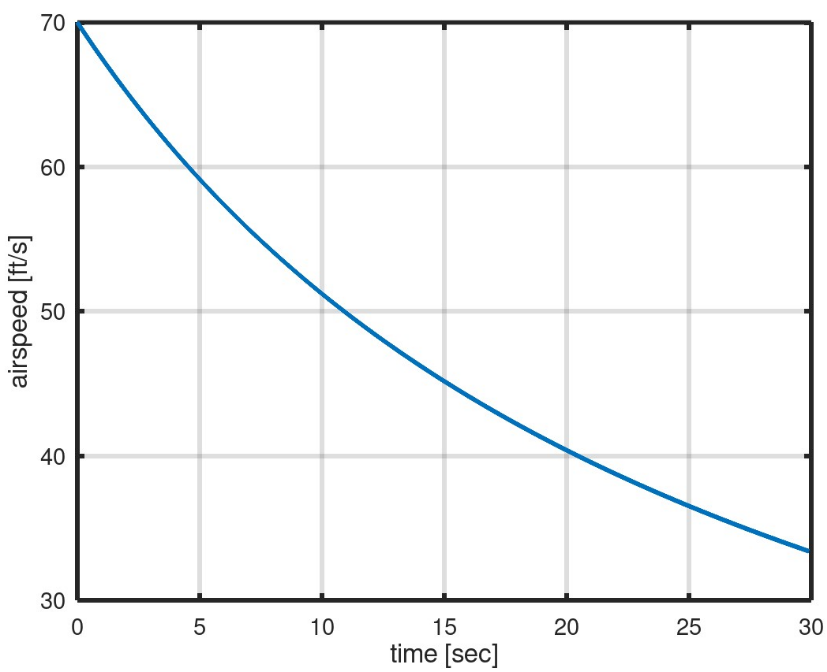
mass, m = 200 slugs
density, rho = 0.002377 slug/ft^3
airship volume, ∇ = 89,300 ft^3
drag coefficient, CD = 0.044

airship velocity after coasting 30 seconds = ________________ [ft/s]

MATLAB/OCTAVE

```
% Problem 1.1

clear % clear old

% Airship Equations
function xdot = EOM(x)
    m = 200.0; % mass [slugs]
    rho = 0.002377; % density [slug/ft3]
    vol23 = 89300.0^(2/3); % volume^(2/3) [ft2]
    CD = 0.044; % drag coefficient []

    tt = x(1); % time [sec]
    V = x(2);  % airspeed [ft/s]

    dV_dt = -0.5*rho*V*V*vol23*CD/m; % airspeed rate [ft/s2]

    xdot = zeros(2,1);
    xdot(1) = 1.0; % time [sec]
    xdot(2) = dV_dt; % dV_dt output
end

% Euler Integration
function xnew = Euler(xold, h)
    xdot = EOM(xold);
    xnew = xold + h*xdot;
end
```

```
% simulation setup
dt = 0.1;  % integration time step [s]
nsteps = 300; % number of time steps

% set initial conditions
x = [0.0; % start time [s]
70.0]; % airspeed [ft/s]

% run simulation
for i = 1:nsteps
    xs(:,i) = x; % store state vector in columns
    x = Euler(x, dt); % Euler step
end

% extract variables for plotting
times = xs(1, :); % time vector [s]
Vs = xs(2, :); % airspeed [ft/s]

% plot output
plot(times,Vs,"linewidth",3.0)
set(gca,"linewidth",3.0,"fontsize",14);
xlabel("time [sec]","fontsize",16);
ylabel("airspeed [ft/s]", "fontsize",16);
grid on;
```



ANSWER: airship velocity after coasting 30 seconds = 33 [ft/s]

Problem 1.2

Two Helios spacecraft were deployed in the mid 1970’s to study solar processes of the sun. Each spacecraft had a mass of about 370 kg and both were designed to spin at a constant 60 rev/min about the 3-axis. Simulate the spacecraft’s initial condition response with 1-axis rotation rate (omega_1) of 0.1 rad/s and the 2-axis rotational rate (omega_2) of 0.01 rad/s. Use the Ralston integration method with step size of 0.05 seconds and run the simulation for a total elapsed time of 30 seconds. Submit an image plot (PNG, BMP, JPG) with 1-axis rotation rate (units: deg/s) on the x-axis and the 2-axis rotational rate (units: deg) on the y-axis ordinate.

I_1 = 182 kg.m^2 I_2 = 187 kg.m^2 I_3 = 194 kg.m^2

$$I_1 \frac{d\omega_1}{dt} = I_1 \dot{\omega}_1 = (I_2 - I_3)\,\omega_3 \omega_2$$

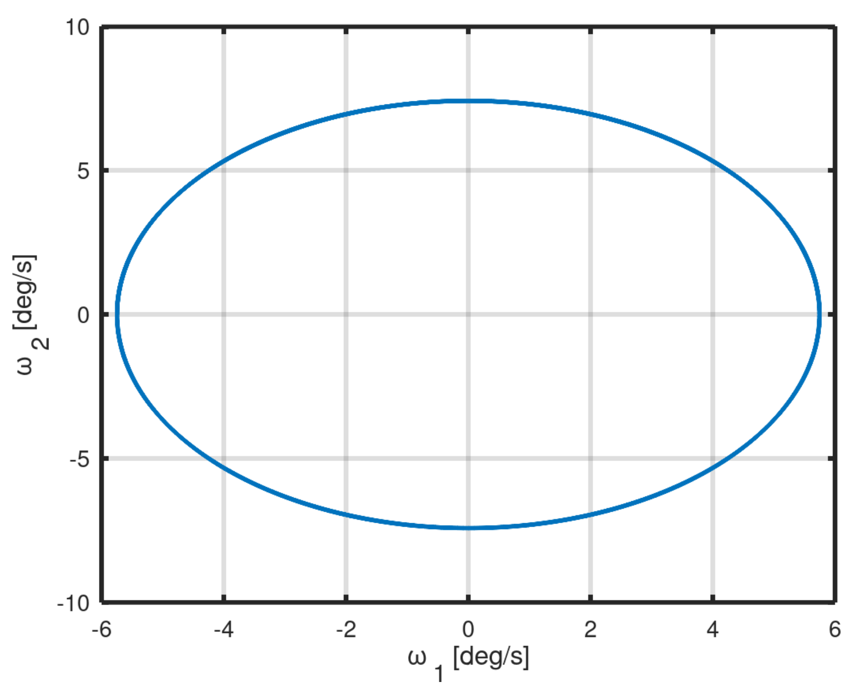
$$I_2 \frac{d\omega_2}{dt} = I_2 \dot{\omega}_2 = (I_3 - I_1)\,\omega_3 \omega_1$$

$$I_3 \frac{d\omega_3}{dt} = I_3 \dot{\omega}_3 = 0$$

MATLAB

```
% Helios Simulation (MATLAB/OCTAVE)
clear  % clear old

% Helios Equations of Motion
function xdot = EOM(x)

    time = x(1); % time [s]
    w1 = x(2);  % 1-axis rotation [rad/s]
    w2 = x(3);  % 2-axis [rad/s]
    w3 = x(4);  % 3-axis [rad/s]

    im1 = 182;
    im2 = 187;
    im3 = 194;

    w1_dot = (im2-im3)*w3*w2/im1;
    w2_dot = (im3-im1)*w3*w1/im2;

    xdot = zeros(3, 1);
    xdot(1) = 1.0; % time [s]
    xdot(2) = w1_dot; % w1_dot [rad/s/s]
    xdot(3) = w2_dot; % w2_dot [rad/s/s]
    xdot(4) = 0.0; % w3 constant
end

% Ralston Integration
```

```
function xnew = Ralston(xold, h)
    r1 = EOM(xold);
    r2 = EOM(xold + (2.0/3.0)*h*r1);
    xnew = xold + 0.25*h*(r1 + 3.0*r2);
end

% simulation setup
dt = 0.05;  % integration time step [s]
nsteps = 600; % number of time steps

% set initial conditions
x = [0.0; % start time [s]
0.1; % w1 [rad/s]
0.01;  % w2 [rad/s]
2.0*pi]; % w3 [rad/s]

% run simulation
for i = 1:nsteps
    xs(:,i) = x; % store state vector in columns
    x = Ralston(x, dt); % Ralston step
end

% extract variables for plotting
times = xs(1, :); % time vector [s]
w1s = (180/pi)*xs(2,:);
w2s = (180/pi)*xs(3,:);

% plot output
plot(w1s,w2s,"linewidth",3.0)
set(gca,"linewidth",3.0,"fontsize",14);
xlabel('\omega _1 [deg/s]',"fontsize",16);
ylabel('\omega _2 [deg/s]', "fontsize",16);
grid on;
```



Problem 1.3

Alaska Airlines recently introduced non-stop flights between San Diego CA (KSAN) and Honolulu HI (PHNL). Estimate the total distance between KSAN and PHNL using two different modes of navigation. The first mode assumes the airplane travels along a rhumb line path between each of the waypoints listed below. The second navigation mode assumes the pilot ignores the intermediate waypoints and follows a great circle path between KSAN and PHNL. Report as your answer the magnitude of the difference between the total trip distance computed by each mode of navigation. Ignore aircraft altitude and assume a spherical Earth with semi-major axis equal to 6378137 m.

Name	Latitude (deg)	Longitude (deg)
KSAN	32.7335556	-117.1896667
ETECO	30.2858333	-131.6361111
DIALO	28.7016667	-139.4266667
DUSAC	26.0061111	-146.1675000
DRAYK	23.0241667	-152.5777778
PHNL	21.3178275	-157.9202627

magnitude of trip distance difference = _________________ [km]

MATLAB/OCTAVE

```
% Problem 1.3

clear

% great circle distance
function dgc = great(A, B)
    Re = 6378137; % semi-major axis
    d2r = pi/180.0; % convert deg to radians
    latA = d2r*A(1);
    lonA = d2r*A(2);
    latB = d2r*B(1);
    lonB = d2r*B(2);
    dgc = Re*acos(sin(latA)*sin(latB)+cos(latA)*cos(latB)*cos(lonB - lonA));
end

% rhumb line distance
function drh = rhumb(A, B)
    Re = 6378137; % semi-major axis
    d2r = pi/180.0; % convert deg to radians
    latA = d2r*A(1);
    lonA = d2r*A(2);
    latB = d2r*B(1);
    lonB = d2r*B(2);
    tauA = log(tan(latA) + 1.0/cos(latA));
    tauB = log(tan(latB) + 1.0/cos(latB));
    psi = atan2(lonB - lonA, tauB - tauA);
    drh = Re*abs(latB-latA)*abs(1.0/cos(psi));
end
```

```
% waypoints
KSAN = [32.7335556 -117.1896667];
ETECO = [30.2858333 -131.6361111];
DIALO = [28.7016667 -139.4266667];
DUSAC = [26.0061111 -146.1675000];
DRAYK = [23.0241667 -152.5777778];
PHNL =      [21.3178275 -157.9202627];

% rhumb line track
rdist = rhumb(KSAN, ETECO);
rdist = rdist + rhumb(ETECO, DIALO);
rdist = rdist + rhumb(DIALO, DUSAC);
rdist = rdist + rhumb(DUSAC, DRAYK);
rdist = rdist + rhumb(DRAYK, PHNL);

% direct great circle path
gdist = great(KSAN, PHNL);

diff = 0.001*(rdist - gdist); % difference [km]
```

ANSWER: distance difference = 10 km

Problem 1.4

A private pilot plans to fly a rhumb line course from San Diego airport (KSAN) to San Francisco airport (KSFO). If there is no wind, the pilot expects the trip to take about 130 minutes at a airspeed of 180 nmi/hr. However, when the actual flight occurs, the wind is blowing (from west) toward east at 20 nmi/hr. How long will the flight actually take if the pilot maintains the 180 nmi/hr airspeed? (1 nmi = 6076 ft)

Name	Latitude (deg)	Longitude (deg)
KSAN	32.7336	-117.1897
KSFO	37.6167	-122.3750

a. 120 minutes

b. 130 minutes

c. 140 minutes

d. 160 minutes

MATLAB/OCTAVE

```
% Problem 1.4

clear

% rhumb line distance and heading
function [drh, psi] = rhumb(A, B)
    Re = 6378137;
    d2r = pi/180.0; % convert deg to radians
    latA = d2r*A(1);
    lonA = d2r*A(2);
    latB = d2r*B(1);
    lonB = d2r*B(2);
    tauA = log(tan(latA) + 1.0/cos(latA));
    tauB = log(tan(latB) + 1.0/cos(latB));
    psi = atan2(lonB - lonA, tauB - tauA); % course angle A to B [rad]
    drh = Re*abs(latB-latA)*abs(1.0/cos(psi)); % distance A to B [m]
end

KSAN = [32.7336 -117.1897];
KSFO = [37.6167 -122.3750];

[rd_m, chiG] = rhumb(KSAN, KSFO); % distance between airports [m]
rd_nmi = rd_m/0.3048/6076; % convert distance to [nmi]

V = 180.0; % airspeed [nmi/hr]
WE = 20.0; % east wind [nmi/hr]
WN = 0.0; % north wind [nmi/hr]

chi = chiG + asin(WN*sin(chiG)/V - WE*cos(chiG)/V); % flight path heading [rad]
VG = V*cos(chi-chiG) + WE*sin(chiG) + WN*cos(chiG); % speed over ground [nmi/hr]

etime = 60.0*rd_nmi/V; % expected time no wind [minutes]
atime = 60.0*rd_nmi/VG; % actual time with wind [minutes]
```

ANSWER: atime = 140 [minutes]

Problem 1.5

Estimate the ratio of atmospheric density at Coors Field in Denver CO (altitude = 5280 ft), on a non-standard “hot” day, relative to the sea-level or base level for the same day.

density ratio = rho/rho1 = rho(5280)/rho(0)

a. 0.83 c. 1.17

b. 0.86 d. 1.20

SOLUTION

From lecture slides (slide 5 “Atmosphere”):

T1 = 560 deg R
h1 = 0 ft

a = lapse rate = (450 deg R – 500 deg R)/((30 – 15)(1000 ft))

a = (-3.33 deg R)/(1000 ft)

T = T1 + a*(h – h1)

T = 560 deg R + (-3.33 deg R)*(5280 ft)/(1000 ft)

T = 542 deg R

For gradient region:

$$\frac{\rho}{\rho_1}=\left(\frac{T}{T_1}\right)^{-[1+g/a/R]}$$

1 + g/a/R = 1 + [32.2 (ft/s2)]*[(1000 ft)/(-3.33 deg R)]*[(slug.deg R)/(1716 ft.lb)]*[(lb.s2)/(slug.ft)]

1 + g/a/R = -4.635

rho/rho1 = [(542 deg R)/(560 deg R)]^(+4.635)

ANSWER: rho/rho1 = 0.86